Stage VI-0.2: Hierarchical World Model v2

Frozen-data offline smoke validation

2026-07-12

1 Decision

Decision D - insufficient frozen artifacts is the single main decision.

2 Artifact gate

The 131-field frozen exact schema contains no fixed-k spectrum, eigenvalue curve, DOS proxy, spectral-minimum count, or spectral curvature. Free-energy basin metadata were not misused as spectral fingerprints.

3 Executed smoke

Levels 1, 2, and 4 were revalidated. Level 3 compares the old log-gap/k-gap output with a log-gap-only ablation. Within-run, cross-run, calibration, risk-coverage, probe, replay, and reconstruction tests were completed.

4 Numerical summary

Mean spatial SC AUROC is 0.977. Median spatial R^2 is 0.432 (Level 1), 0.799 (Level 2), 0.086 (log-gap-only Level 3), and 0.908 (Level 4). No run/level replay uncertainty trajectory is monotonic.

5 Conclusion

The log-gap-only ablation is not a replacement spectral fingerprint. Full-local and EIG are blocked. No learned state is transferable.

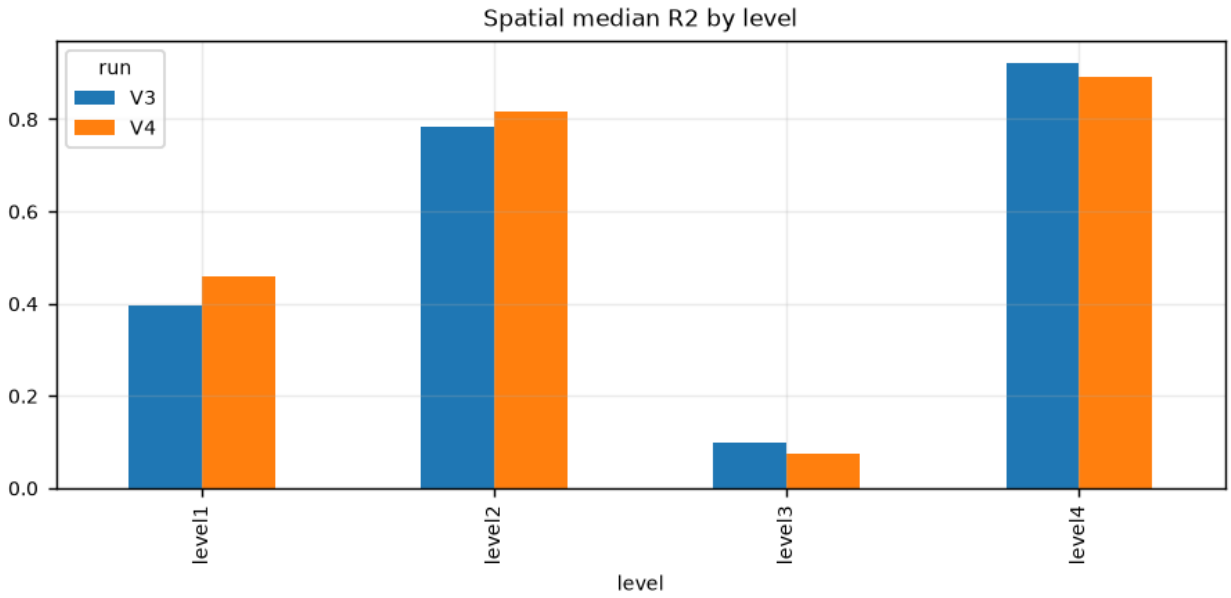


Figure 1: figure 01 spatial r2

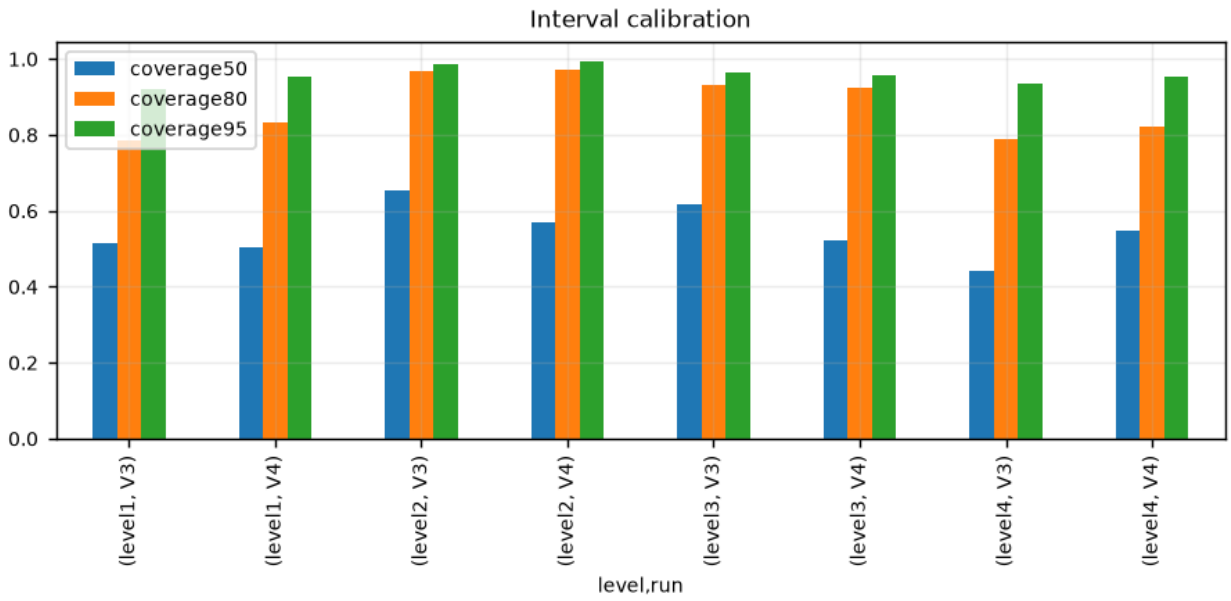


Figure 2: figure 02 calibration

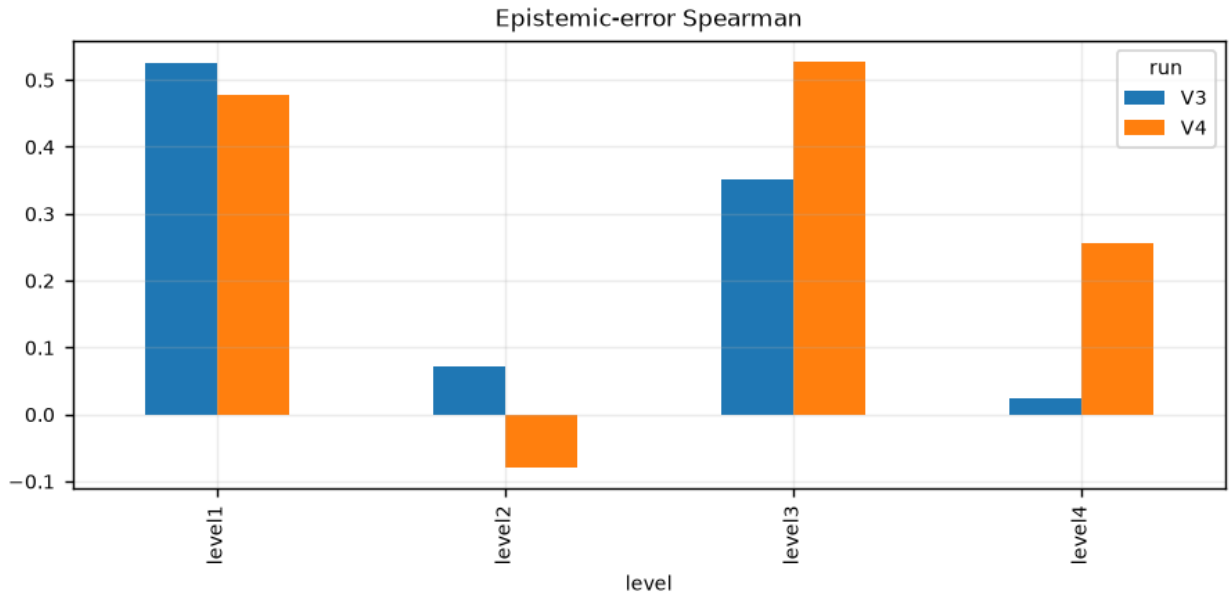


Figure 3: figure 03 uncertainty error

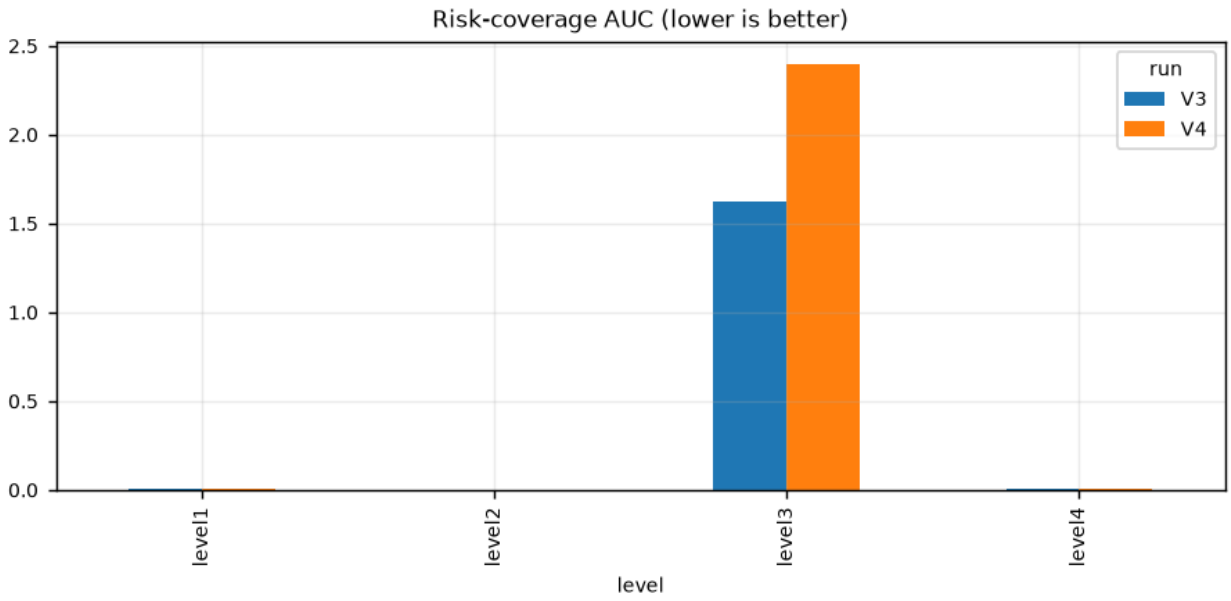


Figure 4: figure 04 risk coverage

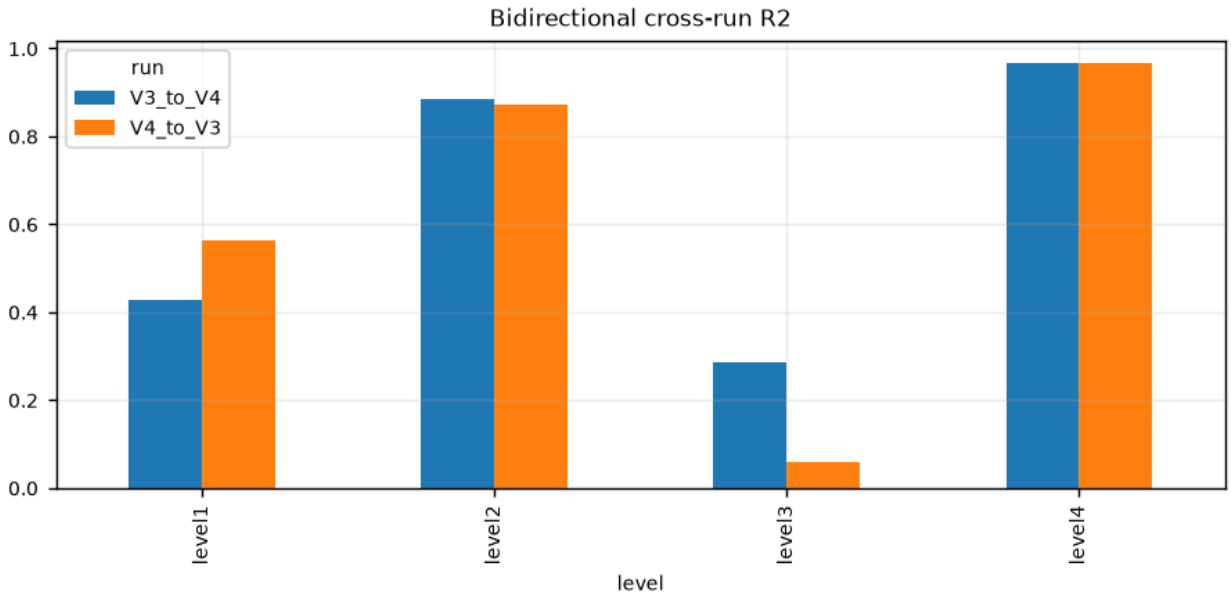


Figure 5: figure 05 cross run

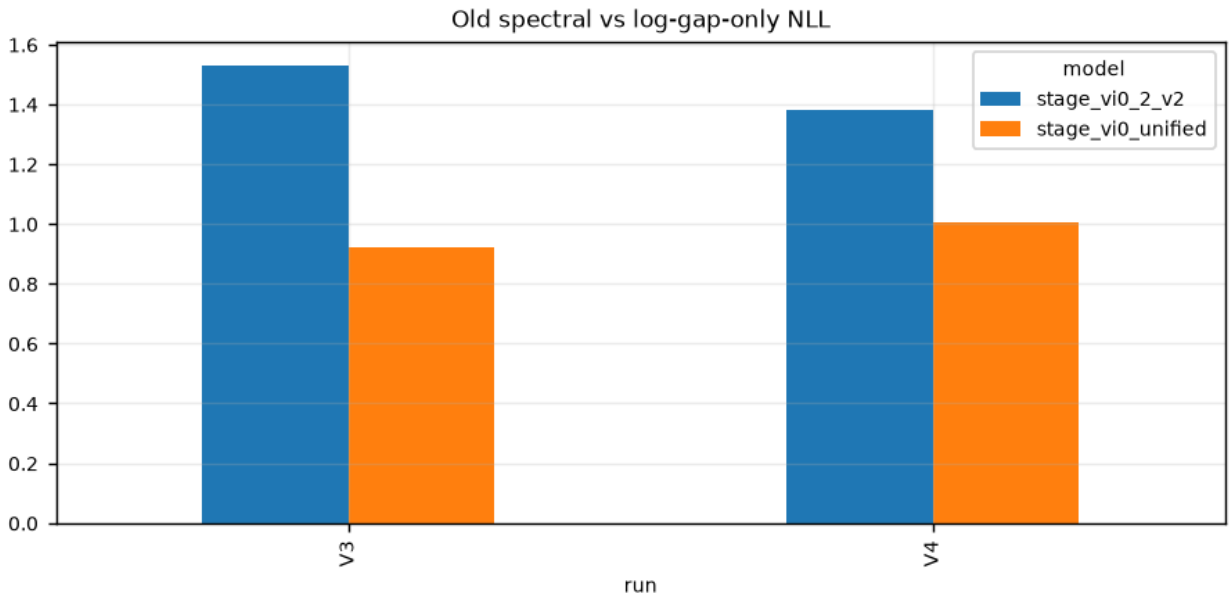


Figure 6: figure 06 spectral comparison

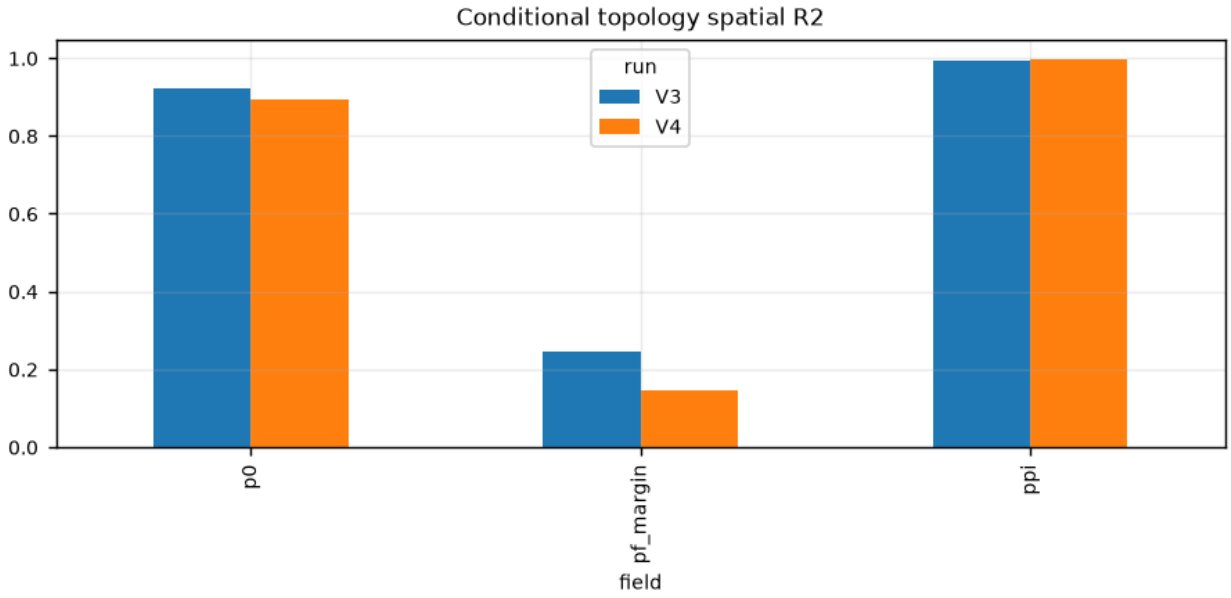


Figure 7: figure 07 topology metrics

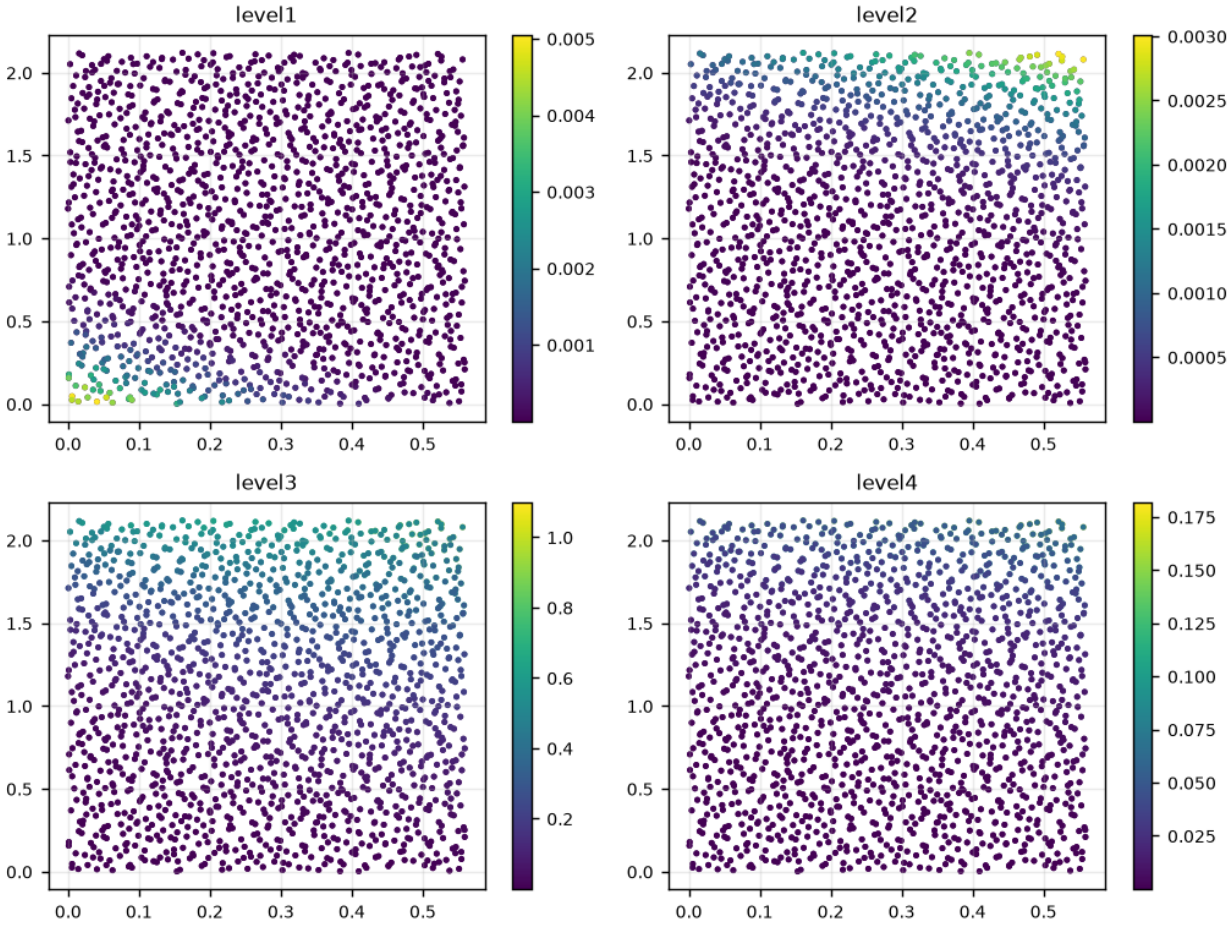


Figure 8: figure 08 probe levels

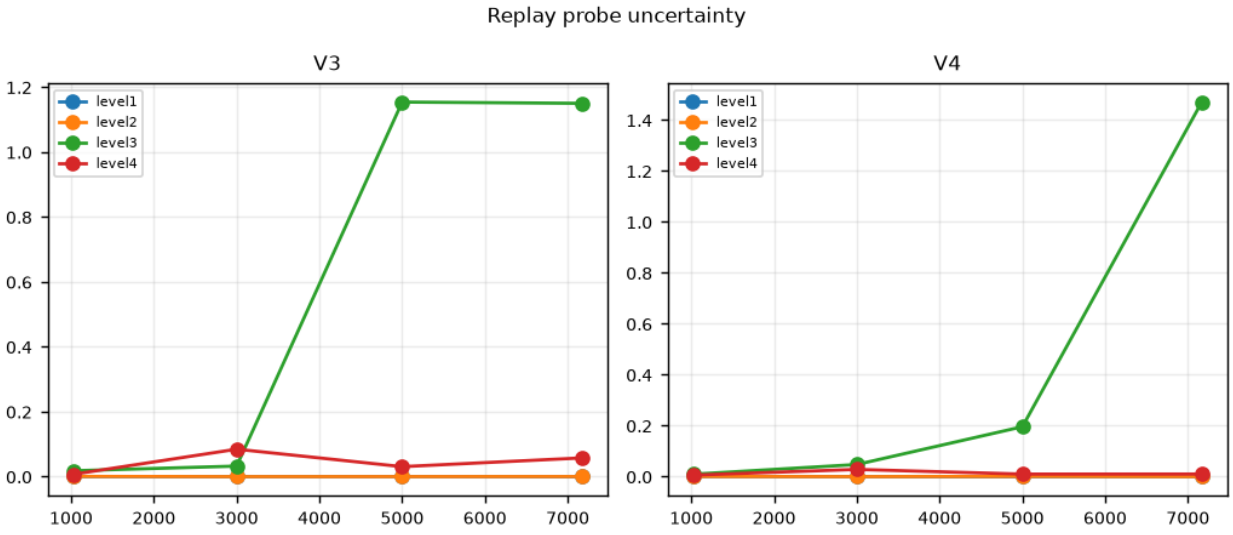


Figure 9: figure 09 replay uncertainty

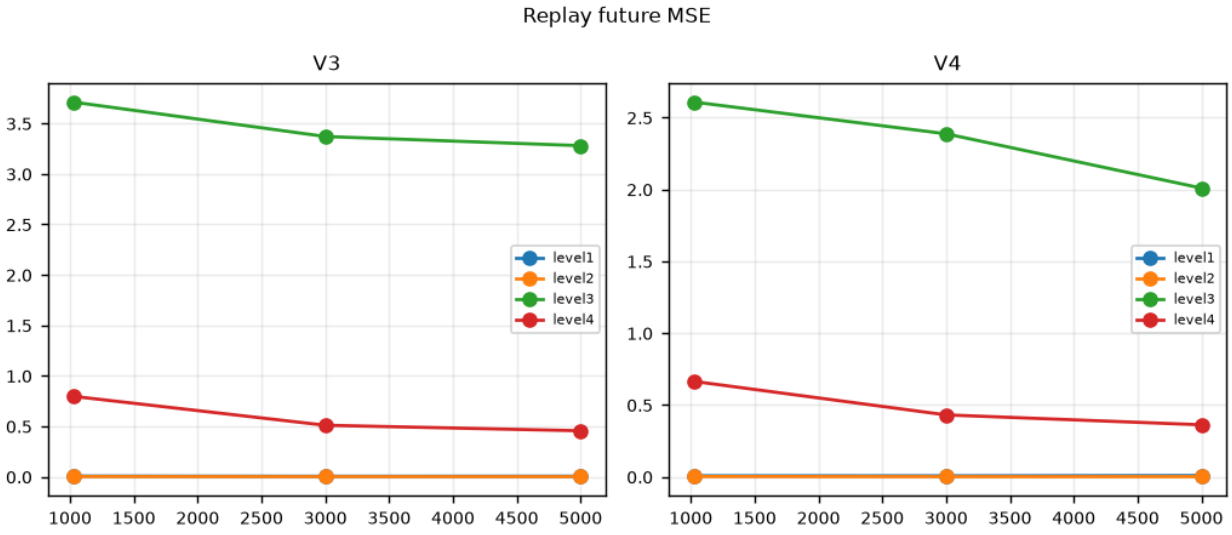


Figure 10: figure 10 replay future

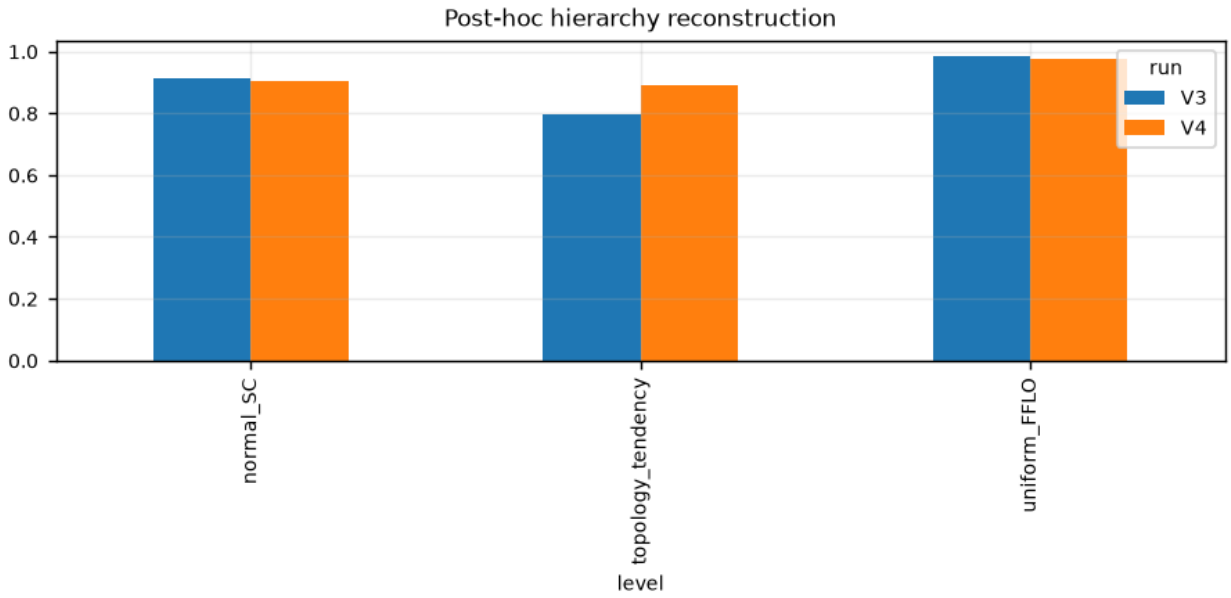


Figure 11: figure 11 phase reconstruction

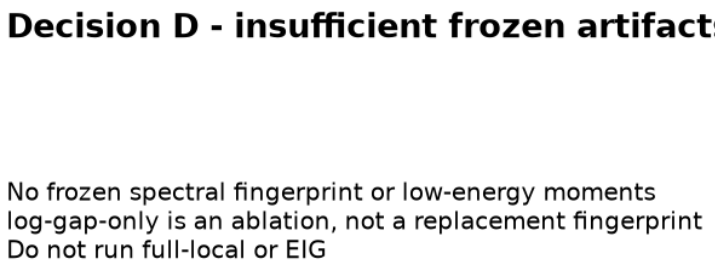


Figure 12: figure 12 decision

6 Next artifact action

Export pre-existing fixed-grid spectra or physically defined low-energy moments if available. This report does not authorize exact calculations.